
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Despite strong empirical performance, generative image models trained on large-scale corpora can fail to instantiate apparently elementary concepts—e.g., human hands or objects appearing in groups of four—that should plausibly be represented within the training distribution. Such failures have primarily been reported through anecdotal evidence, so it remains unclear whether they constitute isolated generation artifacts or systematic limitations in the learned generative distribution. We address this gap by introducing a systematic procedure for identifying and characterizing "conceptual blindspots"—concepts that are present in the training data but are absent, suppressed, or distorted in samples from the model. The method uses sparse autoencoders (SAEs) to estimate interpretable concept embeddings, which provide a common representation for quantifying concept prevalence in real and generated images. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—thereby supporting fine-grained measurement of concept-level discrepancies. When applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the procedure identifies specific suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) as well as exaggerated blindspots (e.g., wood background texture and palm trees). At the individual-datapoint level, the same representation isolates memorization artifacts—cases in which a model reproduces highly specific visual templates observed during training. Overall, we provide a theoretically grounded framework for measuring conceptual fidelity in generative models relative to the underlying data-generating process and for systematically identifying the associated conceptual blindspots.

1 Introduction

Generative image models trained on large-scale datasets now provide high-capacity image synthesis mechanisms, enabling applications within computer vision [1–9] and in downstream domains that use visual generation as a component [9–15]. Despite these capabilities, qualitative [16–18] and quantitative studies [19–20] show that such models can fail to synthesize images containing comparatively simple concepts, including human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. When prompts request these concepts, the generated sample may contain visually related structures without realizing the target concept precisely; a typical example is a hand rendered with six fingers. We refer to these failure modes as "conceptual blindspots" [1]. They are technically nontrivial because the relevant concepts plausibly occur often enough in training data to be encoded by the model. This observation motivates a

¹We borrow the term "blindspots" from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision's rationale.



Figure 1: $\delta(k)$ measures whether model samples over- or under-represent concept c_k relative to its frequency in natural data. Concepts satisfying $\delta(k) < 0.1$ are classified as **suppressed conceptual blindspots**, whereas concepts satisfying $\delta(k) > 0.9$ are classified as **exaggerated conceptual blindspots**. The generated examples from four generative image models illustrate suppressed, exaggerated, and aligned concepts.

36 distributional question: do these failures correspond to idiosyncratic defects for a few concepts, or
 37 do they instantiate a broader phenomenon in which many concepts are generated with substantially
 38 lower or higher probability than under the data-generating process?

39 Existing evaluation protocols for generative image models do not directly resolve this question [28]
 40 [29]. Many standard metrics are coarse-grained statistics designed to assess image realism, such
 41 as FID [30], and therefore cannot localize concept-specific distributional errors. Metrics such as
 42 CLIPScore and distribution-coverage statistics [31–33] provide partial information about diversity
 43 or coverage, but they operate above the granularity of fine-grained features and concepts [34, 35];
 44 consequently, they are not sufficient for identifying conceptual blindspots. Qualitative evaluation
 45 procedures, including participant surveys [5, 17, 36–38] and open-ended model exploration [39],
 46 can reveal concept failures, but they do not define a scalable and repeatable protocol for comparing
 47 concept-level behavior across model families.

48 **This work.** We argue that identifying conceptual blindspots requires an automated and unsupervised
 49 methodology that estimates which concepts in the data distribution have mismatched occurrence
 50 probabilities under the true data-generating process and under the trained generator. Our contributions
 51 in this work are three-fold:

- 52 • **Formalizing Conceptual Blindspots in Generative Image Models.** We define a framework for
 53 identifying and quantifying conceptual blindspots in generative image models relative to natural
 54 images (Section 3). The formalism replaces anecdotal or manually specified concept evaluation
 55 with a principled criterion for measuring model limitations.
- 56 • **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse
 57 Autoencoders.** We specify an SAE-based procedure for extracting sparse concept representations
 58 and comparing the induced concept distributions for natural and synthesized images (Section 4).
 59 As the representation backbone, we train and open-source an archetypal SAE (RA-SAE) on
 60 DINOv2 features with 32,000 concepts, the largest such RA-SAE to date.
- 61 • **Exploratory Tool and Analysis.** We provide an exploratory web tool for distribution-level and
 62 datapoint-level analysis of blindspots across models (Sections 5.1–5.4). We apply the method
 63 to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5), identifying suppressed
 64 conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents)
 65 and exaggerated conceptual blindspots (e.g., wood background texture and palm trees),
 66 as illustrated in Fig. 1.

67 The model, code, and web interface are released through a software repository.

68 2 Related Work

69 2.1 Explainability in Vision

70 Early explainable-AI methods for computer vision primarily estimated attribution over input re-
 71 gions [40–42]. These methods provide limited semantic characterization of internal representations
 72 and can produce unreliable explanations [43, 45]. Concept-based interpretability [46] addresses this
 73 limitation by identifying semantically meaningful directions in neural networks, thereby characteriz-
 74 ing not only spatial evidence but also the concepts and structures encoded by the model [47–49].

Recent analyses show that several concept-based interpretability methods—ACE [44], CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—can be viewed as variants of a dictionary-learning objective with different constraints [52]. Among these methods, SAEs provide a scalable mechanism for concept-based representation analysis. Although standard SAEs exhibit limitations such as overly specific features [53], compositionality failures [54], and limited causal intervention effects [55], subsequent variants, including archetypal SAE (RA-SAE) [56], impose additional structure on the learned dictionary.

Beyond SAEs, prompt-based probing [57], attention-map or activation visualization [58, 59], and dataset-level statistics [32, 33] (e.g., diversity or distribution-coverage metrics) provide only partial evidence for concept-level failure analysis. Their main limitation is representational granularity: they typically operate on complete images or prompts rather than on fine-grained features and concepts [34, 35]. Moreover, they often require subjective interpretation and do not separate distinct classes of model failure [60]. Consequently, existing methods and metrics are insufficient for systematic identification of feature-level and concept-level weaknesses in generative image models [28].

2.2 Generative Image Models

Diffusion-based methods now constitute a dominant modeling family across generative vision tasks, including image [4, 61–64], video [65–72], and 3D synthesis [7, 73–75]. For image generation, this lineage begins with denoising diffusion probabilistic models (DDPMs) [76] and extends to non-Markov diffusion processes such as denoising diffusion implicit models (DDIMs) [77].

The Stable Diffusion (SD) [9] family substantially increased access to DDPM-based image generation in both research and open-source settings. The original SD model was followed by SD 2 [78], SD 3 [79], and SD XL [80]. Subsequent extensions impose additional constraints on the diffusion process, including style [81–82], pose [83], and identity [84, 85], and support additional input modalities such as image-to-image generation. Related latent-diffusion systems built around this paradigm include Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89].

2.3 Datasets for Generative Image Models

The empirical success of generative vision models depends heavily on computational scale and large internet-derived datasets [90–92]. LAION-5B [93] is particularly important for open-source text-to-image training pipelines, including SD and its derivatives. The dataset is scraped from Common Crawl [94] and contains more than 5B image-caption pairs, including 2.3B English pairs. Other large-scale resources include COYO-700M [95] and Conceptual Captions [96], with 700M and 3M image-caption pairs, respectively.

The use of LAION-5B also raises documented concerns about dataset bias and harmful content [97–101]. Filtering procedures did not remove all harmful material [97–99], including NSFW content [100] and hundreds of CSAM instances [101], which led to temporary removal from official channels. The dataset further contains low-quality images [102] and internet-style captions, such as product descriptions, that may be misaligned with user prompt distributions [103].

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning estimates sparse representations in which each sample is reconstructed as a linear combination of a small number of dictionary atoms [104–106]. Building on compressed-sensing theory [107, 108], the field developed from vector quantization [109] to methods such as Non-negative Matrix Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. Recent work studies online optimization [114], structured sparsity [115], theoretical recovery guarantees [116, 117], and connections to deep neural representations [118, 119].

Sparse-coding methods have also been used to analyze the emergence of high-level concepts in diffusion models [120]. Before diffusion models, concept-grounded interpretability was applied to generative architectures through concept-bottleneck models, which require human intervention during training [121], and post-hoc detectors that retrofit concept supervision [122]. Both approaches depend on human-specified concept sets and therefore cannot detect broad concept-level trends that are not enumerated in advance.

Aggregate metrics, including precision, recall, density, and coverage [35, 123], together with latent-density scores that estimate sample quality from the model latent space [124], provide distribution-level evaluation of generative image models. These metrics can reveal distributional gaps, but they do not specify which concepts are under- or over-represented.

These limitations motivate a scalable, unsupervised framework for identifying and quantifying concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We first formalize *conceptual blindspots* as systematic discrepancies between the conceptual content induced by the natural image distribution and the conceptual content induced by model-generated samples. This formulation yields quantitative criteria for determining which concepts are under- or over-represented by a generative model relative to its data distribution. The corresponding construction is illustrated in Fig. 2. Although the formalism uses standard assumptions from representation-learning theory [126–130], our empirical analysis indicates that the proposed measurements expose meaningful phenomena even when these assumptions hold only approximately.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G: \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational economy, we use $p(\cdot)$ for both the latent density $p(c)$ and its pushforward density on image space $p(x)$, with $x = G(c)$. The invertibility of G justifies this convention because it induces a well-defined distribution on $\mathcal{X}$ through the change-of-variables formula. Under this model, individual latent dimensions correspond to the **Concepts** that parameterize the data distribution $P_{\mathcal{X}}$ over the image observation space $\mathcal{X}$. For example, distinct latent coordinates may encode factors such as color, shape, size, or location [131, 132]. We additionally allow the data-generating process to associate a text description $t \in \mathcal{T}$ with each sampled image, while leaving the caption-generation mechanism implicit. These text descriptions provide conditioning variables for training a text-conditioned **Generative image model** g_{θ} , with parameters θ , on image-text pairs so that a noise sample $\eta \sim \mathcal{N}(0, I)$ and a scene description t are mapped to an image x that realizes the described content.

To define conceptual blindspots for g_{θ} , we compare the probability mass assigned to a concept by the data-generating process with the corresponding mass induced by the model distribution. Accordingly, we start from an evaluation set of natural image-text pairs $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$ and construct a synthetic image set $\mathcal{D}'_{\mathcal{X}}$ by generating from the associated text descriptions. To estimate concept occurrence probabilities, we introduce an **Energy model** $\xi: \mathcal{X} \rightarrow \mathbb{R}^d$ that maps an image to a d -dimensional vector whose k^{th} coordinate is the scalar energy assigned to concept k , i.e., $\xi_k(x) = E(c_k)$. These estimates are well-defined under the assumed invertibility of the data-generating process. The probability assigned to image x can therefore be written as $p_k(x) \propto \exp(-\sum_k \xi_k(x))$, with Z_{ξ} as the partition function, which induces the population-level estimate $p_k(\mathcal{D}_{\mathcal{X}}) = \prod_{x \in \mathcal{D}_{\mathcal{X}}} p_k(x)$. Using this construction and the sigmoid map $\sigma(\cdot)$, we define the average energy difference for concept k between $\mathcal{D}_{\mathcal{X}}$ and $\mathcal{D}'_{\mathcal{X}}$.

Definition 2 (Energy Difference). Let $x \in \mathcal{D}_{\mathcal{X}}$ denote a real image sampled from the data-generating process G , and let $x' \in \mathcal{D}'_{\mathcal{X}}$ be a synthetic image generated by the model g_{θ} . Let $\xi_k: \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the

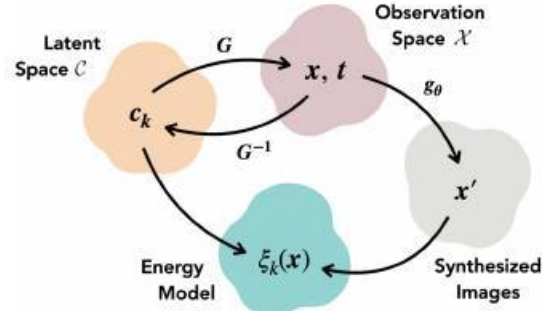


Figure 2: **Mapping induced by the framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) by the invertible DGP. S_{θ} generates x' conditionally on t . The energy model $\xi_k(x)$ estimates concept representations for both x and x' , enabling comparison of the induced concept distributions and identification of blindspots.

179 energy difference for concept k as:

$$\begin{aligned}\delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_{\mathcal{X}})}{p_k(\mathcal{D}_{\mathcal{X}}) + p_k(\mathcal{D}'_{\mathcal{X}})},\end{aligned}\quad (1)$$

180 where the expectations are taken over $\mathcal{D}'_{\mathcal{X}}$ and $\mathcal{D}_{\mathcal{X}}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$
181 denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

182 Thus, the energy difference for concept k represents the relative probability mass of that concept in a
183 target observation set (here, $\mathcal{D}'_{\mathcal{X}}$) compared with a baseline set (here, $\mathcal{D}_{\mathcal{X}}$). This measure induces the
184 following definition of conceptual blindspots.

185 **Definition 3 (Suppressed / Exaggerated Conceptual Blindspots).** Given a generative image model
186 g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in
187 the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

188 In this terminology, a conceptual blindspot is a concept whose occurrence probability in generated
189 images differs substantially from its prevalence under the data-generating process, either through
190 suppression or exaggeration. Suppressed concepts have disproportionately low activation (e.g.,
191 $\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are over-represented (e.g., $\delta(k) > \lambda_{\max}$). In all
192 experiments, we set $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to operationalize these two regimes.

193 This definition is related to the mode-collapse phenomenon studied in prior work (e.g., see Bau
194 et al. [39]), but differs in the level of analysis. Mode collapse concerns exaggerated or suppressed
195 probability of generating *entire images*, whereas our criterion measures probability shifts for indi-
196 vidual concepts. For example, if a model systematically fails to generate an object with a white
197 background, we classify that concept as a suppressed conceptual blindspot.

198 4 Method: Operationalizing the Definition of Conceptual Blindspots

199 We now specify the operational pipeline used to estimate conceptual blindspots for a generative model
200 g_θ . Following Sec. 3, the required objects are (i) images sampled from g_θ under conditions matched
201 to the natural data, and (ii) an energy model that maps both natural and generated images into a
202 comparable concept-energy representation. Throughout this section, $\|\cdot\|_F$ denotes the Frobenius
203 norm and $\|\cdot\|_0$ denotes the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix
204 $\mathbf{X}$, the relation $\mathbf{X} \geq 0$ denotes element-wise non-negativity. For $n > 0$, we define $[n] := \{1, \dots, n\}$
205 and denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

206 **From Prompts to Latent Representations.** To estimate conceptual blindspots for g_θ , we compare
207 a dataset $\mathcal{D}_{\mathcal{X}}$ of image-caption pairs $(\mathbf{x}, \mathbf{t})$ sampled from the data-generating process G with synthetic
208 counterparts generated by conditioning g_θ on the same text descriptions.

209 For each caption $\mathbf{t}$, we synthesize a counterpart image $\mathbf{x}'$ with a pretrained text-to-image generator
210 $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$ implemented as a denoising diffusion probabilistic model (DDPM) [78, 86, 87].
211 Sampling is performed in latent space by a reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\gamma_t, \mathbf{t}, \mathbf{c}) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

212 where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ are specified by the standard cosine noise schedule. The
213 terminal latent γ_0 is decoded by a pretrained VAE to obtain $\mathbf{x}' = \text{VAE}(\gamma_0)$. In the remaining
214 analysis, we abstract away this internal sampler and treat g_θ as a black-box conditional map $\mathbf{t} \mapsto \mathbf{x}'$.

215 **Defining the Energy Model.** Prior work indicates that self-supervised representation learning
216 can invert aspects of the data-generating process and recover the underlying energy function up
217 to linear transformations [126, 128, 133, 134]. We therefore use DINOv2 [135], a state-of-the-art
218 self-supervised vision encoder, as the feature extractor for our analysis. Because the number of
219 concepts underlying the DGP may exceed the dimensionality of the encoder feature space [51, 136],
220 we train sparse autoencoders (SAEs) on DINOv2 features to recover sparse subspaces associated with
221 those concepts [50, 56, 137, 139]. The operative assumption is that if DINOv2 represents generative
222 concepts as approximately orthogonal directions, consistent with the independence condition in Def. 1

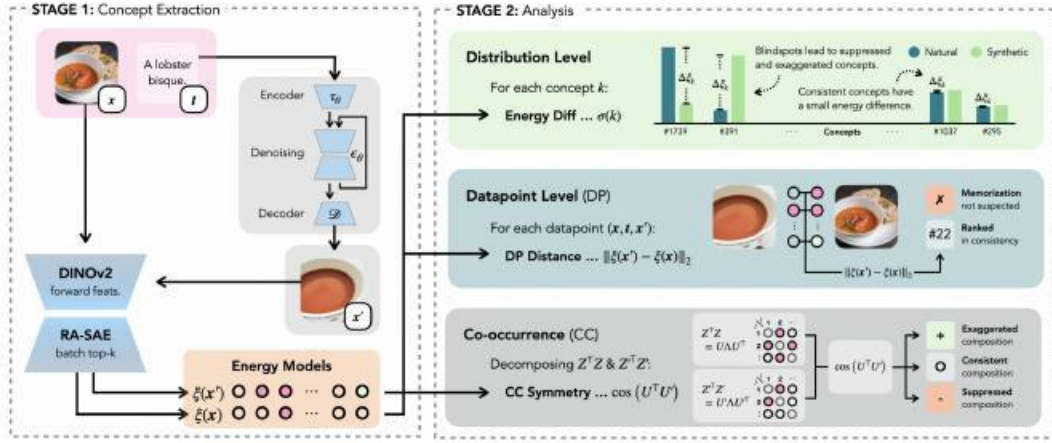


Figure 3: **Concept Extraction Pipeline.** Given a triplet (x, t, x') , we compute DINOv2 features for each image and pass them through a RA-SAE to obtain sparse concept embeddings. These embeddings instantiate the energy representations $\xi(x)$ and $\xi(x')$; in particular, $\xi_k(x) = E(c_k)$ denotes the energy assigned to the k -th concept.

Formally, let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the DINOv2 feature extractor. For each natural or synthetic image x in $\mathcal{D}_{\mathcal{X}}$ or $\mathcal{D}'_{\mathcal{X}}$, we compute $a = f(x) \in \mathbb{R}^d$. For n image-text pairs, the corresponding real and generated feature vectors are stacked into matrices $A, A' \in \mathbb{R}^{n \times d}$. We decompose each feature vector into a sparse linear combination of learned concept atoms with an SAE. Specifically, let $D \in \mathbb{R}^{d \times K'}$ be a dictionary containing K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the SAE encoder that maps input features to sparse codes. Applying Ψ row-wise produces the activation matrix $Z = \Psi(A) \in \mathbb{R}^{n \times K'}$, where row $z_i = \Psi(a_i)$ is the sparse concept decomposition for image i . The decoder reconstructs feature vectors as ZD^T , and training minimizes reconstruction error subject to sparsity and non-negativity constraints:

$$\min_{\Psi, D} \|A - \Psi(A)D^T\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Standard SAEs can converge to arbitrarily oriented dictionaries, which makes downstream concept analysis sensitive to random initialization. To reduce this instability and improve reproducibility, we use the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary D so that each atom is a convex combination of training examples. Specifically, we parameterize $D = WA$ with $W \in \Omega_{K', n}$, where $\Omega_{K', n}$ is the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W\mathbf{1} = \mathbf{1}\}. \quad (3)$$

Consequently, every atom D_i lies in the convex hull of the data $\text{conv}(A)$, and every reconstruction ZD^T lies in the conic hull of the data $\text{cone}(A)$. This constraint keeps the learned concepts anchored to the support of the data distribution [56]. After training, the SAE yields aligned sparse codes: Z for real images and Z' for their generated counterparts. *These codes express prompt-conditioned visual semantics in a shared interpretable basis, and each concept activation functions as the energy value used in our blindspot analysis.* Thus, given a prompt and its associated natural image (t, x) , the pipeline produces two sparse concept vectors (z, z') that support direct comparison between natural and generated visual content in a common conceptual basis.

The complete pipeline therefore starts from a captioned image (t, x) , generates a synthetic counterpart x' , and maps both images through a vision encoder and trained SAE into a shared sparse concept space. The resulting representations (z, z') form the basis for estimating conceptual shifts induced by the generative process.

5 Results

We evaluate four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—using $|\mathcal{X}| = 10,000$ image-text pairs and

the corresponding model generations (Fig. 4). The evaluation is organized into three analytical levels (Fig. 3): a **distribution-level** analysis estimates suppressed and exaggerated concepts; a **datapoint-level** analysis identifies failures associated with ambiguity, omission, and memorization; and a **compositional** analysis characterizes distortions in concept co-occurrence geometry.

A central artifact of the evaluation is the interactive exploratory tool shown in Fig. 5. The tool implements a web-based interface centered on a UMAP projection of concept representations, enabling inspection and comparison of concept-level energy differences. It is released through a software repository together with pre-computed energy-difference data for the four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). The analyses below are derived from the tool’s following functionality:

- **Contrast different models and architectures.** For each evaluated model, the interface renders a UMAP visualization over all 32,000 RA-SAE concepts. Each scatter point corresponds to one concept and is color-coded by its energy difference.
- **Inspect concepts.** Each concept is associated with a detail card containing summary statistics, representative real and generated images (x , x'), and visualized co-occurrence structure.
- **Explore blindspots.** In addition to the UMAP and per-concept views, the interface provides global rankings of suppressed and exaggerated blindspots, which prioritize concepts with the largest measured deviations.

Rather than performing an exhaustive analysis of a single phenomenon, we report representative findings that demonstrate the tool’s applicability and support broader user-directed exploration.

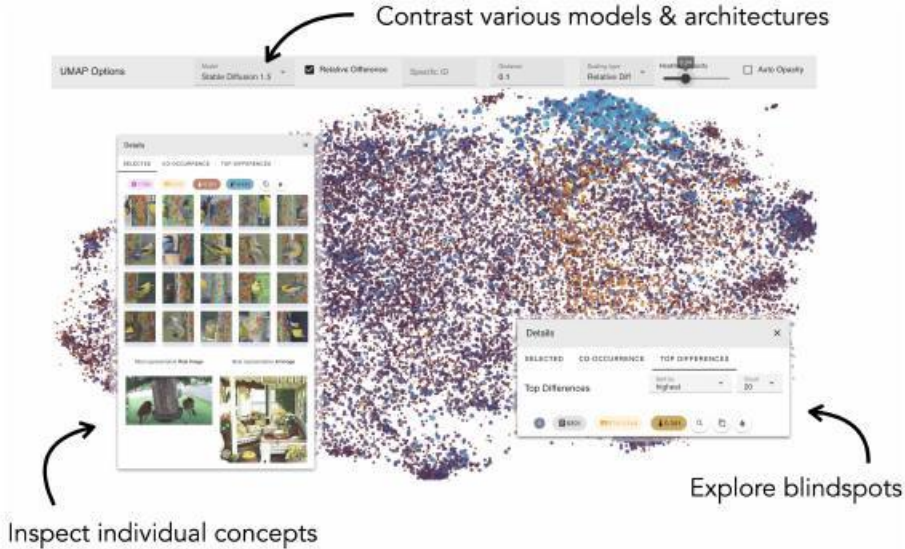


Figure 5: **Overview of the Exploratory Tool.** For each evaluated model, the web interface displays a UMAP projection in which each point denotes one concept and color encodes its energy difference. Selecting a concept opens a detail panel with illustrative images, summary statistics, and representative natural and generated images x and x' . The interface also reports ordered concept co-occurrences and global blindspot rankings.

5.1 Generative Image Models Cannot Produce Unseen Concepts

To quantify discrepancies between each model distribution and the natural image distribution, we first analyze the marginal energy difference $\delta(k)$ for the 32,000 RA-SAE concepts. As defined in Sec. 4

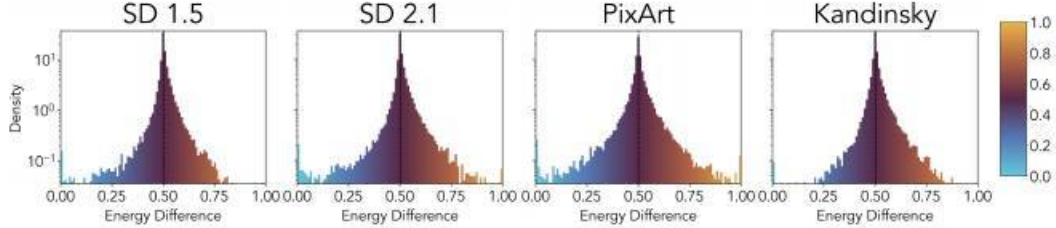


Figure 6: **Concept Energy Distribution.** Log-scale histograms of energy differences $\delta(k)$ over 32,000 concepts for each evaluated model, comparing natural and synthesized concept distributions. Values left of zero correspond to suppressed concepts (under-representation), whereas values right of zero correspond to exaggerated concepts (over-representation).

this statistic measures the relative prevalence of concept k in synthesized images compared with natural images. Values satisfying $\delta(k) < 0.1$ indicate under-representation (suppression), whereas $\delta(k) > 0.9$ indicates over-representation (exaggeration). Fig. 6 reports the empirical distribution of $\delta(k)$ for each evaluated model. Across models, the histograms are heavy-tailed and place nontrivial mass near both extremes, indicating systematic concept-coverage discrepancies. The left tail, which corresponds to suppressed concepts, is consistently denser and longer than the right tail, indicating a stronger tendency toward concept omission. This asymmetry is reflected by negative distributional skewness: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

Although every evaluated model exhibits both suppressed and exaggerated concepts, the model-specific profiles are distinct. For example, PixArt has a wider distributional spread, consistent with a more suppressed concept profile. The shared left skew nevertheless indicates a common tendency toward concept omission; the structure and specificity of this behavior are analyzed in the following sections.

5.2 Structure and Specificity of Conceptual Blindspots

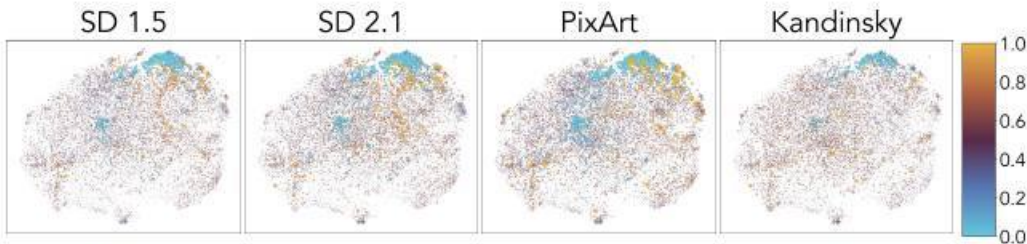


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts, with color indicating the energy difference $\delta(k)$ between natural and synthesized distributions. Spatial clusters identify structured blindspot regions, with suppressed concepts at the blue end of the scale and exaggerated concepts at the yellow end.

Having quantified marginal concept-frequency discrepancies, we next examine their global organization by embedding all 32,000 concepts into two dimensions with UMAP applied to the sparse codes and coloring each concept by its $\delta(\cdot)$ value. Fig. 7 shows that distinct concept clusters appear for all evaluated models. These clusters frequently correspond to contiguous blindspot regions, particularly among suppressed (blue) concepts, which indicates that blindspots have nonrandom structure and may reflect shared properties of training distributions or architectural priors. To evaluate cross-model consistency, we compare both the magnitude and spatial structure of concept-level $\delta(\cdot)$ values. Fig. 8 reports scatter plots and pairwise Pearson correlations between the $\delta(k)$ vector for SD 1.5 and the corresponding vectors for the other models. As expected, SD 1.5 and SD 2.1 are strongly correlated ($r = 0.82$), consistent with their related architectural and training pipelines. Their correlations with PixArt and Kandinsky are lower— $r = 0.41$ and $r = 0.46$, respectively—which indicates that these architectures emphasize different regions of concept space.

Overall, these analyses indicate that some blindspots are shared across models, plausibly due to dataset-level properties, whereas others are model-specific and may arise from differences in training

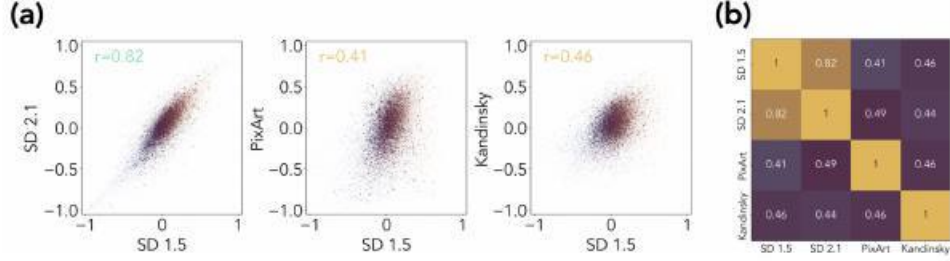


Figure 8: **● Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ for the four evaluated models, with Pearson correlation coefficients reported in the upper-left corner. Strong alignment between SD 1.5 and SD 2.1 contrasts with weaker correlations involving other architectures, indicating model-specific blindspot structure. **● Correlation Matrix of Conceptual Blindspots.** Heatmap of pairwise Pearson correlations for δ across all models, quantifying the extent of shared conceptual blindspots.

315 dynamics or model capacity. The subsequent qualitative analyses therefore distinguish between
 316 blindspots that are common across models and blindspots that are unique to particular models.

317 5.3 Qualitative Examples of Blindspots

318 We next inspect concrete suppressed and exaggerated blindspots to characterize the concepts assigned
 319 to these regimes. In Fig. 9a, we show a suppressed blindspot shared by all models: each evaluated
 320 model fails to reproduce the concept `solid white on documents`. Although the caption explicitly
 321 specifies this concept, none of the generated images realizes the intended visual semantics, suggesting
 322 that this region of concept space is systematically under-sampled across models. Fig. 9b shows a
 323 model-specific blindspot: the concept `pan` is captured by three models but absent from the Kandinsky
 324 generations. This observation is consistent with Sec. 5.2 where some regions exhibit high cross-model
 325 agreement while others show architecture-specific behavior.

326 Conversely, Fig. 10 presents an exaggerated blindspot in which generated images over-emphasize
 327 the concept `shadow under animal`. Although shadows are semantically plausible in these scenes,
 328 their consistent and pronounced rendering across models, compared with the more variable natural-
 329 image instances, suggests an overactive generative prior. Despite searching for model-specific
 330 exaggerated concepts, we did not identify clear examples, suggesting that the observed exaggerations
 331 are approximately shared across models.

332 Together, these examples show how concep-
 333 tual blindspots appear in generated images
 334 and demonstrate that the energy-based diag-
 335 nostic can recover both shared and model-
 336 specific failure modes. In particular, the
 337 method identifies unexpected limitations, such
 338 as the consistent failure to reproduce clear or

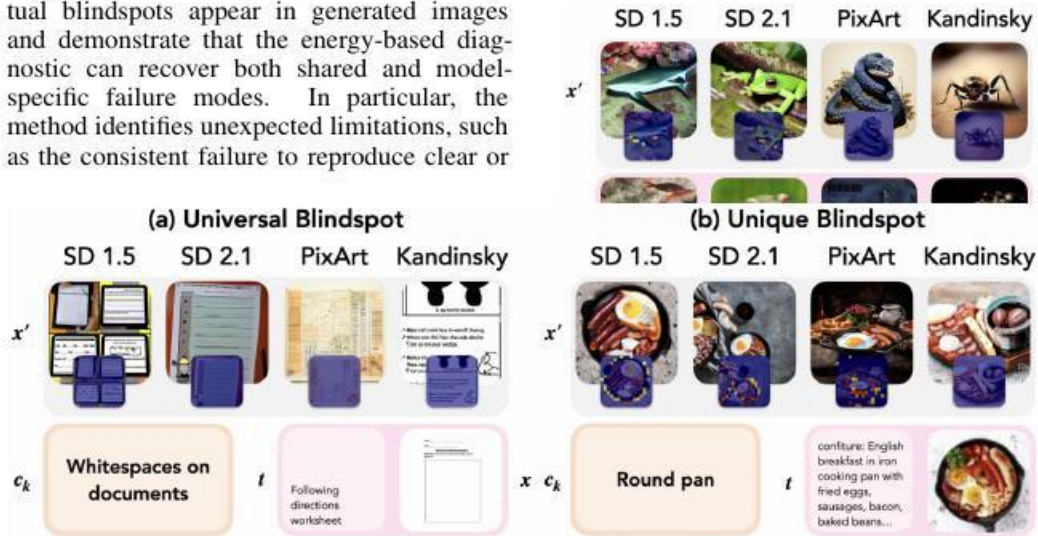


Figure 9: **● Examples of Suppressed Conceptual Blindspots.** Natural images `animal`, `solid white on documents`, and captions t are shown with four synthesized images generated using `SoVITS`. The universal blindspot occurs in all evaluated models, whereas the unique blindspot occurs only in Kandinsky.

solid background regions like whitespaces on documents. Such failures may reflect architectural constraints or training-data biases that are not specific to a single model.

Although aggregate-level analyses are informative, they do not determine whether blindspots arise only as population-level effects or also occur within individual image pairs. A datapoint-level analysis provides a finer diagnostic for mechanisms such as prompt misinterpretation, latent memorization, or their combination.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we examine natural-generated image pairs whose $\delta(\cdot)$ values, averaged over all concepts, attain the largest and smallest differences. This analysis separates apparent model successes from failures that are often attributable to prompt ambiguity or memorization artifacts. Fig. 11a shows instances with near-zero differences in average $\delta(\cdot)$ values. In these cases, the generated images are nearly indistinguishable from the originals in the learned concept representation. Qualitative inspection indicates that this alignment does not necessarily result from faithful prompt-semantic understanding; instead, it can arise from replication of memorized templates. Repeated visual structures, such as clothing outlines or object arrangements, suggest copying from high-frequency training patterns. By contrast, Fig. 11b shows samples with among the largest $\delta(\cdot)$ values. These pairs exhibit substantial conceptual divergence between synthesized and natural images. Some discrepancies are attributable to underspecified or noisy captions, whereas others indicate genuine blindspots: the prompt specifies a clear concept that is present in $\mathcal{D}_{\mathcal{X}}$, but the model does not realize it in $\mathcal{D}'_{\mathcal{X}}$. This pattern suggests that, even when language grounding is adequate, some concepts remain outside the model’s effective generative support.



Figure 11: **• Datapoint-level Conceptual Alignment.** (a) Examples with minimal energy differences, where generated samples appear to reproduce training-pattern templates. (b) Examples with large energy differences, where prompt ambiguity or model limitations induce substantial concept divergence.

5.5 Analyzing Post-Training Effects

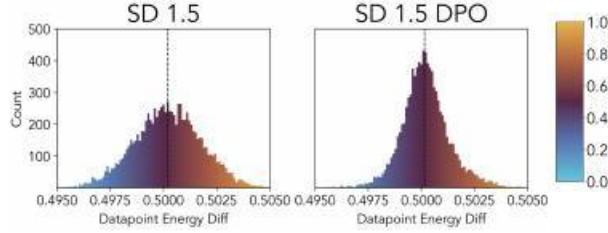


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between synthesized and natural distributions for SD 1.5 variants with and without DPO.

Post-training procedures, including safety fine-tuning, have been argued to reduce the diversity of model generations [140]. Because the proposed pipeline localizes discrepancies between model generations and the ground-truth DGP, we use it to analyze the effect of DPO, a widely used safety fine-tuning protocol [141]. Specifically, we compare two SD 1.5 variants: one trained with DPO and one without DPO. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm between the corresponding concept-energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 reports the histogram of these datapoint-wise energy differences. The DPO-enhanced model has both a lower median and a narrower spread, indicating a generated concept distribution that is more consistently aligned with the ground-truth DGP. This result suggests that DPO may regularize the model’s concept distribution and encourage outputs whose semantics more closely match the observed inputs. Although this analysis does not isolate the specific inductive biases introduced by DPO, the results provide empirical evidence that its human-preference optimization objective indirectly improves alignment with the training distribution. In particular, DPO reduces *both over and under* activation of individual concepts relative to the natural baseline. These findings demonstrate that the pipeline can characterize post-training interventions not only through output quality but also through changes in the conceptual geometry of the model’s output space.

5.6 Conceptual Misalignment as a Function of Empirical Frequency

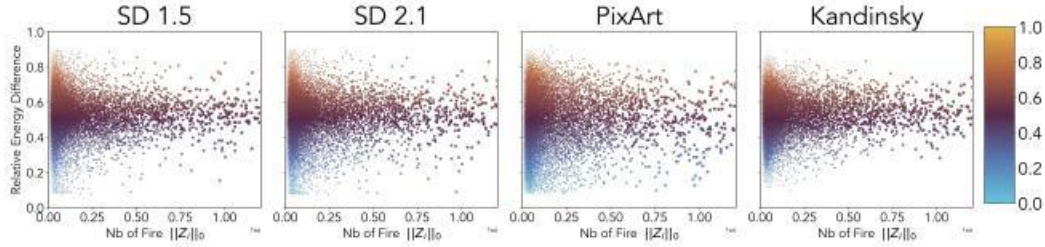


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots of concept frequency (x-axis) versus energy difference (y-axis) for the four evaluated models. Each point denotes one concept, with marker size proportional to activation frequency.

We hypothesized that conceptual blindspots are not solely architectural artifacts, but can also arise from distributional properties of specific features. We test this hypothesis by measuring whether concepts that are rarely activated in natural images, i.e., concepts in the long tail of the data distribution, are more likely to become blindspots in generative models. Concretely, we pass the natural dataset $\mathcal{D}_X$ through the trained SAE and compute, for each concept k , its empirical frequency $\|Z_{:,k}\|_0$, where $Z_{:,k}$ denotes the activations of concept k over all images. We then relate this frequency statistic to the absolute energy difference measured on generated outputs. Fig. 13 visualizes this relationship for each evaluated model. Concepts with higher empirical frequency in natural data tend to have lower energy discrepancies, whereas rare concepts—especially suppressed concepts ($\delta(k) < 0.5$)—exhibit larger alignment errors. This pattern suggests that many blindspots are not random artifacts or isolated model quirks, but systematic effects associated with long-tail concept distributions. Mitigating these failures may therefore require data-centric strategies such as reweighting or augmentation in addition to architectural changes.

6 Discussion

Our analysis identifies multiple conceptual blindspots across four widely used generative image models. The reported results should be interpreted as an initial characterization rather than an exhaustive account: each observed failure mode could support a dedicated follow-up study. We therefore emphasize the generality of the method and exploratory tool, which jointly support systematic identification of hard-to-generate concepts, detection of memorization artifacts, discovery of datapoints with insufficient captions, measurement of post-training effects, and comparison of conceptual shifts across model architectures. These capabilities define several directions for future work, including deeper causal analysis of blindspot origins, evaluation of additional architectures, and hierarchical RA-SAE representations that would support more refined concept analyses.

Limitations. Several limitations qualify the interpretation of our results. Because concept extraction is implemented through DINOv2 and RA-SAE, the analysis is restricted to concepts that are represented by these components; concepts that are weakly encoded by this representation pipeline may not be detected. In addition, although the evaluation set contains 10,000 images, it may not fully cover rare concepts, long-tail co-occurrences, or higher-order compositional statistics.

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Appendix

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A Experimental Setup

This section specifies the experimental configuration used to analyze four generative image models—SD 1.5/2.1, Kandinsky, and PixArt—all trained on LAION-5B or on related subsets and derivatives. The code is released through a software repository.

A.1 Observation Space

The observation space is constructed by sampling 10,000 image-text pairs from LAION-5B [93], which defines the natural-image domain for evaluation. Because the original release is no longer available due to CSAM and other unsafe-content concerns, we use a filtered substitute subset available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure comprises three steps: (1) loading LAION with the Hugging Face datasets library, (2) validating URL structure and resource availability through HTTP HEAD requests, and (3) sampling with replacement until 10,000 valid examples are obtained. This procedure yields D_G as (x, t) tuples and ensures that the evaluation set contains accessible image-text pairs despite the unavailability of many URLs from the original release. Additional synthesized examples are shown in Appendix H.

A.2 Synthesized Images

For each evaluated model, we generate a synthetic dataset D_{g_θ} in one-to-one correspondence with D_G , producing triplets (x, x', t) . For each of the 10,000 pairs (x, t) in D_G , the caption t is used as the conditioning input for model g_θ to synthesize x' .

Synthesis follows the standard text-to-image pipeline for each architecture and is implemented with the Hugging Face diffusers library. Models are loaded at mixed precision (fp16), and all generated images use 512×512 pixel resolution with default generation parameters.

Stable Diffusion 1.5. We use the checkpoint at <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5>, which mirrors the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>. Inference uses 50 denoising steps and guidance scale 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5, used in Sec. 5.5, retains the baseline SD 1.5 pipeline but replaces the UNet with the DPO-trained component from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. We use the checkpoint at <https://huggingface.co/stabilityai/stable-diffusion-2-1>. Inference uses 50 denoising steps and guidance scale 7.5.

Kandinsky. We use the checkpoint at <https://huggingface.co/kandinsky-community/kandinsky-2-1>. Inference uses 100 denoising steps and guidance scale 4.0.

PixArt. We use the checkpoint at <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS>. Inference uses 50 denoising steps and guidance scale 7.5.

A.3 Distribution Level Analysis

Section 5.1 For each evaluated model, we compute $\delta(\cdot)$ for all 32,000 concepts. Normalization applies a sigmoid transformation with temperature $T = 0.8$. The resulting values are plotted as log-scale density histograms with 100 bins over $[0, 1]$.

Section 5.2 We embed all 32,000 concepts into two dimensions by applying UMAP to the sparse concept codes. Each point denotes one concept and is colored by $\delta(\cdot)$, thereby indicating suppressed and exaggerated blindspots. Cross-model consistency is quantified with pairwise Pearson correlations between $\delta(\cdot)$ vectors for all model pairs, yielding both scatter plots and correlation matrices. This analysis tests whether blindspots cluster in concept space and separates model-specific from universal blindspot patterns.

Section 5.3 We rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually inspect the extrema corresponding to suppressed and exaggerated blindspots. Suppressed examples are selected from concepts with $\delta(\cdot) < 0.1$; exaggerated examples are selected from concepts with $\delta(\cdot) < 0.9$. Presented examples are manually annotated with textual descriptions of the respective concepts through inspection of their most activating images and spatial attention patterns. We outline ongoing efforts to automate this concept interpretation in Appendix D

Section 5.5 We compare 1.5 with and without DPO in the following fashion: for each image pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$. We apply a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences before taking their mean. This yields datapoint-wise energy differences that quantify how much each generated image deviates from its natural counterpart in concept space. Finally, these differences are visualized as overlapping histograms, contrasting both model variants.

Section 5.6 For each concept c_k , its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$ (the count of non-zero activations across the natural dataset) is counted. A sigmoid normalization with temperature $T = 0.4$ is then applied to the energy differences $\delta(\cdot)$. The analysis is visualized using scatter plots where the x-axis is the empirical concept frequency and the y-axis is the sigmoid-transformed energy difference. The point sizes are proportional to activation frequency and point colors are proportional to the magnitude of energy differences.

A.4 • Datapoint Level Analysis

Section 5.4 For each image pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$. This yields a scalar measure of conceptual divergence for each image pair. The samples are ranked by their energy differences. Minimal divergence indicate potential memorization artifacts and maximal divergence point to significant conceptual failures. This analysis enables qualitative inspection of specific failure modes.

A.5 • Co-occurrence Analysis

Appendix F For both the natural and synthesized data D_G and D_{g_θ} , concept co-occurrence patterns are analyzed through the co-activation matrix $Z^T Z$, which holds pair-wise correlations in concept usage. Spectral analysis is performed to examine the dominant conceptual directions using eigendecomposition. The alignment between natural and synthetic co-occurrence structures is assessed using cosine similarity heatmaps between the top-100 eigenvectors of each co-occurrence matrix. These 100×100 similarity matrices are visualized as square heatmaps where perfect diagonal alignment would indicate identical principal concept axes, while off-diagonal patterns reveal would revolve rotations and mismatches in compositional geometry.

937 B Custom RA-SAE Training Details

938 This section specifies the training configuration for our custom archetypal SAE (RA-SAE) [56]. The
939 trained model is released through a software repository.

940 **Dataset.** The auto-encoder is trained on the full ImageNet-1k training split, containing ≈ 1.28 M
941 RGB images. Each image is encoded as 261 visual tokens with DINOv2 [135]; these tokens are
942 provided to the SAE without class or position embeddings. The total training-token count is therefore
943 $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

944 **Dictionary.** The dictionary contains 32,000 concept dimensions. Sparsity is imposed by top- k
945 masking with $k = 5$, so all activations outside the largest five for each input are set to 0. We initialize
946 weights with Xavier/Glorot initialization. Training uses mixed precision (fp16), with the final ten
947 epochs run at full precision.

948 **Optimizer and Schedule.** The model is trained for 50 epochs with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
949 and weight decay 10^{-5} . The learning-rate schedule uses linear warm-up for the first 5% of steps,
950 followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The objective combines MSE
951 reconstruction loss with an auxiliary penalty on activations that never enter the top- k set, weighted by
952 $\lambda = 10^{-5}$.

C Computational Resources

This section reports the GPU resources used for training and evaluation. In total, the experiments used approximately 202 GPU-hours on NVIDIA H100 and H200 GPUs.

RA-SAE. Training required approximately 24 GPU-hours distributed across three NVIDIA H100 GPUs.

Synthesized Images. Generating the complete $D_{g\theta}$ (Appendix A) required approximately 5 hours per generator when distributed over four NVIDIA H200 GPUs. Across five generators, this corresponds to approximately 100 NVIDIA H200 GPU-hours.

● **Distribution-Level Analysis.** Distribution-level energy-difference extraction required approximately 3 hours per generator on a single NVIDIA H200 GPU, for a total of ~ 15 GPU-hours.

● **Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts within each datapoint, and ranking datapoints within each concept also required approximately 3 hours per generator on one H200 GPU, for a total of ~ 15 GPU-hours.

D Autointerpretability

To assess whether RA-SAE concepts can be automatically interpreted, we qualitatively evaluated the ability of Vision-Language Models (VLMs) to describe them. We queried ChatGPT 4o through the OpenAI API with multiple prompts and image formats for an image x'' in which a target concept was salient, and recorded the generated description. This section summarizes that exploratory evaluation. During evaluation, we allowed the VLM to return its full reasoning; for scalable deployment, the prompt could be modified to request only the final concept description.

First, we provided the VLM with two inputs: the raw image x'' and a version of x'' with a heatmap overlay indicating concept intensity:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DINOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

In this example, the VLM did not correctly interpret the heatmap. Rather than describing the wood texture in the background, which is the RA-SAE-salient region, it described the foreground ring, which was not emphasized by the RA-SAE.

To test whether the failure was caused by prompt underspecification, we repeated the query with an expanded prompt that described the heatmap semantics in greater detail:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

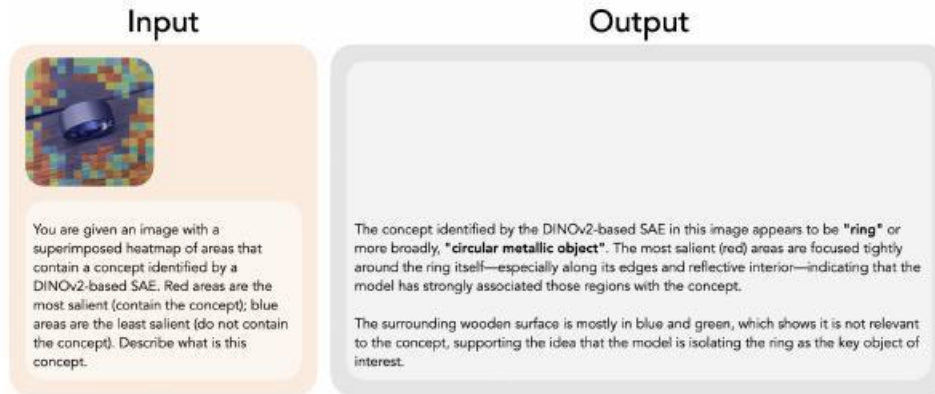
Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

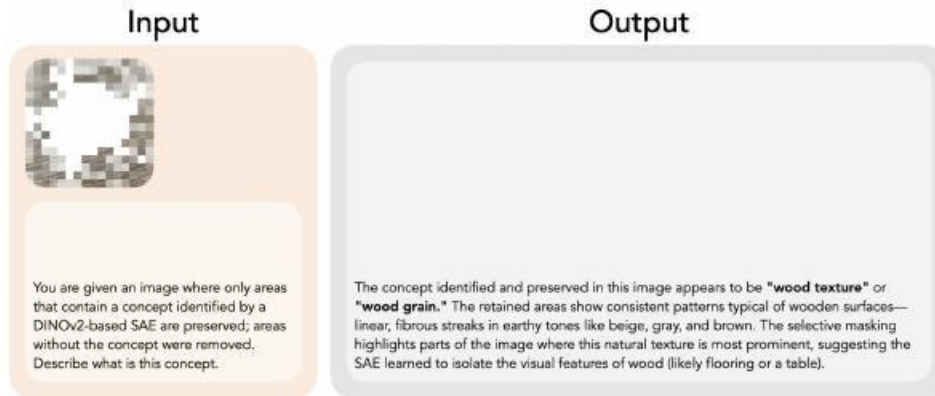
This prompt modification did not correct the behavior: the VLM again identified the ring as the prominent concept. Additional prompting variants, including prompts that described the heatmap construction in greater detail or included the code used to generate it, produced similar outcomes.

983 We next provided only the heatmap-superimposed image x'' , together with a similar prompt explain-
 984 ing the intended heatmap interpretation:



985 Again, the VLM failed to identify the wooden background as the prominent concept and instead
 986 focused on the ring. Additional prompting strategies with this image format produced qualitatively
 987 similar results.

988 These observations indicate that the heatmap overlay alone is insufficient to direct the VLM toward
 989 the RA-SAE-salient regions. Less relevant image regions appear to dominate the VLM description.
 990 To reduce this confounding visual context, we replaced the heatmap-superimposed x'' with an
 991 alpha-masked version of x'' :



992 With this representation, the VLM correctly identified wood texture as the dominant concept in
 993 the example. We observed qualitatively consistent success across multiple concepts and images.

994 The alpha-masked formatting of x'' combined with the prompt above produced the most reliable
 995 behavior in our qualitative evaluation. This evaluation remains limited because it lacks ground-truth
 996 concept annotations and considers only a single VLM. Future work on SAE autointerpretability can
 997 extend this analysis with annotated benchmarks and broader model comparisons.

998 E Additional Results: Model-Specific Blindspots

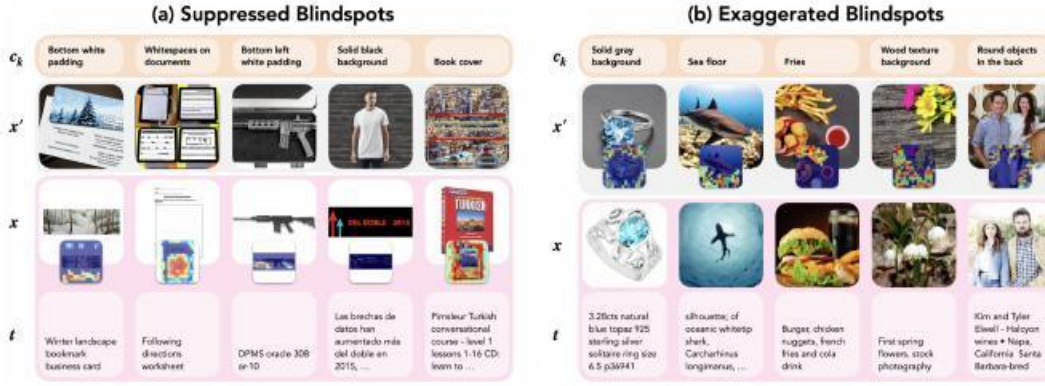


Figure 14: Conceptual blindspot examples for **Stable Diffusion 1.5**. For each concept, the figure shows the prototypical natural image for suppressed blindspots or synthesized image for exaggerated blindspots, selected by highest absolute activation. The concept-specific spatial heatmap is overlaid on the image.

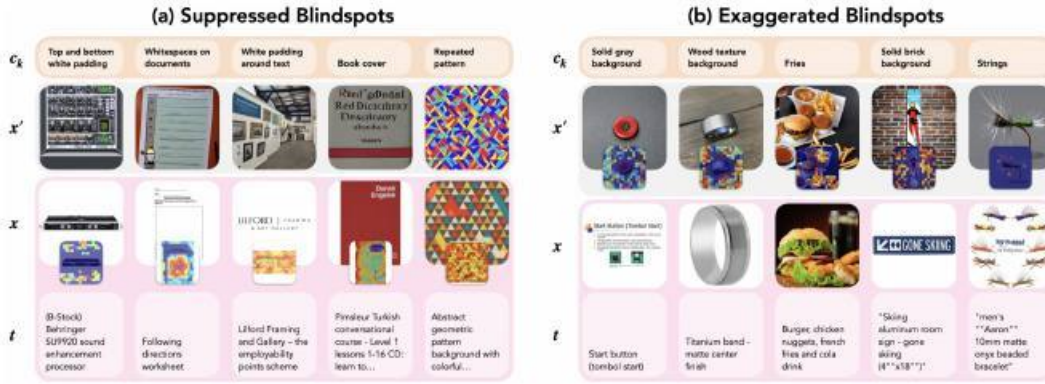


Figure 15: Conceptual blindspot examples for **Stable Diffusion 2.1**. For each concept, the figure shows the prototypical natural image for suppressed blindspots or synthesized image for exaggerated blindspots, selected by highest absolute activation. The concept-specific spatial heatmap is overlaid on the image.

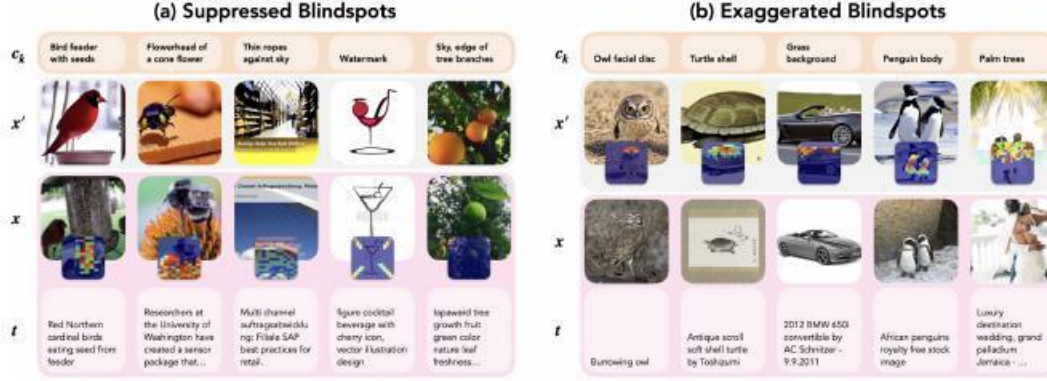


Figure 16: Conceptual blindspot examples for **Kandinsky**. For each concept, the figure shows the prototypical natural image for suppressed blindspots or synthesized image for exaggerated blindspots, selected by highest absolute activation. The concept-specific spatial heatmap is overlaid on the image.



Figure 17: Conceptual blindspot examples for **PixArt**. For each concept, the figure shows the prototypical natural image for suppressed blindspots or synthesized image for exaggerated blindspots, selected by highest absolute activation. The concept-specific spatial heatmap is overlaid on the image.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, the analysis has focused on individual concept activations. Visual scenes, however, are not merely collections of isolated concepts; they also contain co-occurrence structure that encodes compositional semantics. We therefore examine whether generative models preserve this higher-order structure by analyzing the co-activation matrix $Z^T Z$, which measures pairwise correlations in concept usage.

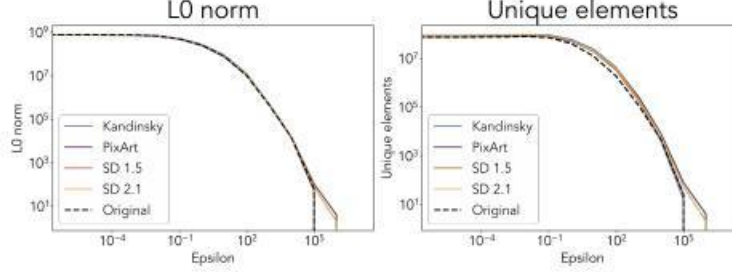


Figure 18: **Sparsity and Structural Divergence.** Left: L_0 norm of the co-occurrence matrix $Z Z^T$ as a function of threshold ϵ , measuring the number of active entries in each model. Right: number of entries that are unique to the synthesized distribution relative to the natural distribution. All evaluated models preserve global sparsity structure but diverge in activation content.

At the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As shown in Fig. 18 (left), the ℓ_0 norm of $Z^T Z$ —thresholded at different ϵ values—closely matches between natural and synthesized distributions across all models. This indicates that the coarse connectivity of the conceptual graph, i.e., which concepts co-activate at nonzero frequency, is largely preserved. Formally, $Z^T Z$ can be interpreted as the adjacency matrix of a weighted undirected graph over concepts, with edge weights given by dataset-level co-activation strength.

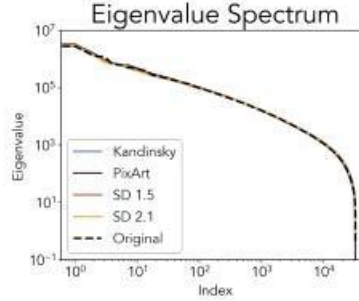


Figure 19: **Spectral Structure of Co-occurrence.** Log-log eigenvalue spectra of co-occurrence matrices $Z Z^T$ for each model and for the natural distribution. All evaluated models match the heavy-tailed decay observed in the natural distribution.

However, Fig. 18 (right) shows that the content of the co-activations diverges: a substantial fraction of entries in model-generated $Z^T Z$ is not shared with the natural baseline. Thus, although global compositional capacity is retained, the identities of active concept pairings can shift, potentially reflecting model-specific inductive biases or training artifacts. To characterize the internal structure of these co-occurrence patterns, we next perform spectral analysis. Fig. 19 reports the eigenvalue spectra of the co-occurrence matrices for each model and for the natural distribution. All spectra exhibit heavy-tailed decay consistent with power-law behavior, indicating that the models preserve the overall rank structure and variance allocation across concept dimensions.

We further measure alignment of dominant conceptual directions using cosine-similarity heatmaps between the top 100 eigenvectors of synthesized and natural co-occurrence matrices (Fig. 20). All models exhibit partial diagonal alignment, indicating overlap in principal concept axes; however,

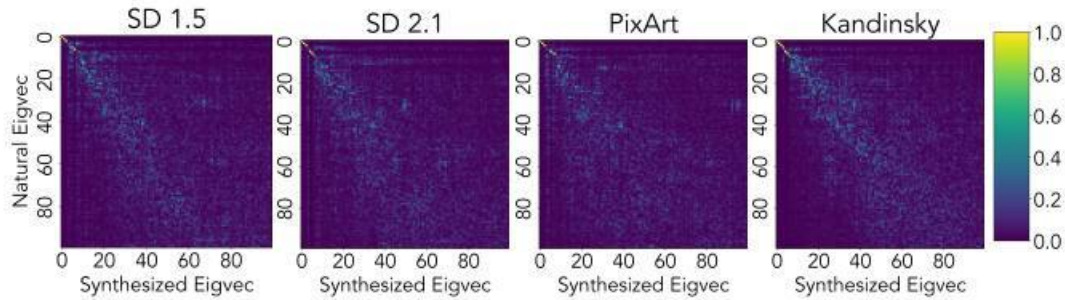


Figure 20: ● **Concept Basis Similarity.** Cosine-similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure indicates alignment of dominant conceptual directions, with model-dependent alignment strength.

1025 off-diagonal entries reveal rotations and mismatches in higher modes, evidencing deviations in
 1026 compositional geometry.

1027 Together, these findings indicate that diffusion models approximate the global structure of concept
 1028 co-activation while retaining structured deviations that become visible under spectral analysis. These
 1029 higher-order discrepancies may contribute to failures in generating coherent multi-object scenes or
 1030 relational concepts.

1031 G Additional Results: Blindspot Stress Testing

1032 To stress-test blindspots identified by the method, we collected prompts that explicitly describe each
 1033 target blindspot and used them to generate multiple images. We then compared outputs from models
 1034 for which the target concept was classified as a blindspot against outputs from models for which it
 1035 was not.

1036 Specifically, ChatGPT-4o was queried with the following prompt: *I want to generate an image of the*
 1037 *following concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input*
 1038 *for a text-to-image model. Return these as a list of strings in Python.* For each prompt, we generated
 1039 five images and analyzed them with our custom RA-SAE model (Appendix B), which ranked samples
 1040 by the activation intensity of the target concept. We then manually reviewed all images to determine
 1041 whether the blindspot concept was successfully rendered.

1042 The suppressed-concept examples below show that models occasionally reproduce partial attributes
 1043 of the target concept, such as a holder or string for bird feeder and a circular hole for glossy
 1044 DVD disc. However, they generally fail to render the complete concept, which is consistent with the
 1045 method's blindspot classification and supports the stress-test validity.

1046 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Stress-test generations for prompts involving the bird feeder concept. For Kandinsky, the method classifies this concept as a suppressed conceptual blindspot, matching the observed failure to generate a corresponding object. In contrast, SD 1.5 is not classified as blindspot-positive for this concept and generates the concept successfully.



Figure 22: Stress-test generations for prompts involving the glossy DVD disc concept. For SD 1.5, the method classifies this concept as a suppressed conceptual blindspot, matching the observed failure to generate a corresponding image. In contrast, Kandinsky is not classified as blindspot-positive for this concept and generates it successfully.

1048 **H Additional Examples: Synthesized Images**



Figure 23: Additional LAION-5B image-caption examples with prompt-matched generations from SD 1.5, SD 2.1, PixArt, and Kandinsky.

1049 I Concentration bounds for δ

1050 In our experiments, we estimate $\delta(k)$ using $n = 10,000$ paired samples for each concept. While this
 1051 budget is modest, it raises the natural question of whether it suffices to obtain reliable estimates. To
 1052 address this, we derive a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ using McDiarmid's
 1053 inequality [142]. The resulting bound is tight and demonstrates that even with relatively few samples,
 1054 we can obtain fast and accurate estimates of concept bias.

1055 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1056 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1057 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1058 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1059 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1060 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1061 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1062 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1063 By McDiarmid's inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1064 which proves the claim. □

1065 Practically, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only
 1066 a few reaching values up to 10. The concentration bound shows that even for the largest observed
 1067 biases, a sample size of $n = 10,000$ yields estimates of $\hat{\delta}_n(k)$ that deviate from the true value by
 1068 no more than a small ε with high probability. This justifies our sampling strategy and confirms that
 1069 accurate bias measurements are attainable with limited data.

J Monotonicity and Calibration-Free Interpretation of δ

Our goal when analyzing blind spots is to rank concepts by the severity of their generative bias. In practice, we use the score $\delta(k)$ for this purpose. However, one may wonder whether such a score introduces distortions relative to more direct quantities such as the energy gap or the odds ratio. The following result establishes that $\delta(k)$ is a strictly increasing reparameterization of both, and therefore inherits their ordering. This guarantees that no calibration is needed when using $\delta(k)$ to rank concepts.

Theorem 5 (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of $\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.

Proof. The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict monotonic functions never reverse inequalities, the three quantities share the same total order over concepts.

□

Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by energy gap or by conceptual generation odds ρ_k . No calibration is necessary, and all three quantities preserve the same total ordering over concepts.

K Stability of FID Under SAE Embeddings

In this section we establish a quantitative relationship between the Fréchet Inception Distance (FID) computed in the original activation space (of dimension d) and the FID after applying a (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout we assume that D has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We first use the following elementary case: if D is square and orthogonal with $k = d$, then the map between A and Z is an isometry, so FID is exactly preserved. Because this case is not realistic for overcomplete SAEs, we then consider the overcomplete setting and bound the FID distortion using the extremal singular values of D . We work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is $\mathcal{W}_2^2$ specialized to Gaussian distributions.

For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically, $\sigma_{\min}, \sigma_{\max} \approx 1$ in our setting, but the proof does not assume this. We begin with a lemma for the non-overcomplete case.

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D preserves the Euclidean norm. Since push-forward by D applies the same map inside the $\mathcal{W}_2$ optimization, the transport integrand is unchanged, and therefore the infimum has the same value. $\square$

This exact isometry case is not representative of typical SAEs, which usually rely on overcompleteness to extract meaningful and interpretable concepts. In the overcomplete case, D is not square orthonormal, but it can still be column-orthonormal. We use this condition to bound distortion through the extremal singular values associated with $D^\top D$.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1127 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1128 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1129 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1130 Using $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and substituting $(\mathbf{A}, \mathbf{A}')$ (respectively,
 1131 $(\mathbf{Z}, \mathbf{Z}')$) completes the proof. $\square$

1132 Theorem 6 states that applying a column-orthogonal overcomplete SAE dictionary can distort
 1133 Fréchet Inception Distance by at most the square of its extremal singular values. When $\mathbf{D}$ is nearly
 1134 orthogonal—as observed empirically with $\sigma_{\min}, \sigma_{\max} \approx 1$ —FID measured in the SAE feature space
 1135 remains close to canonical FID. $\square$

1136